Week 5: Resampling Methods

DATA 201: STATISTICAL METHODS FOR DATA SCIENCE

COURSE SHELL

SPRING 2026

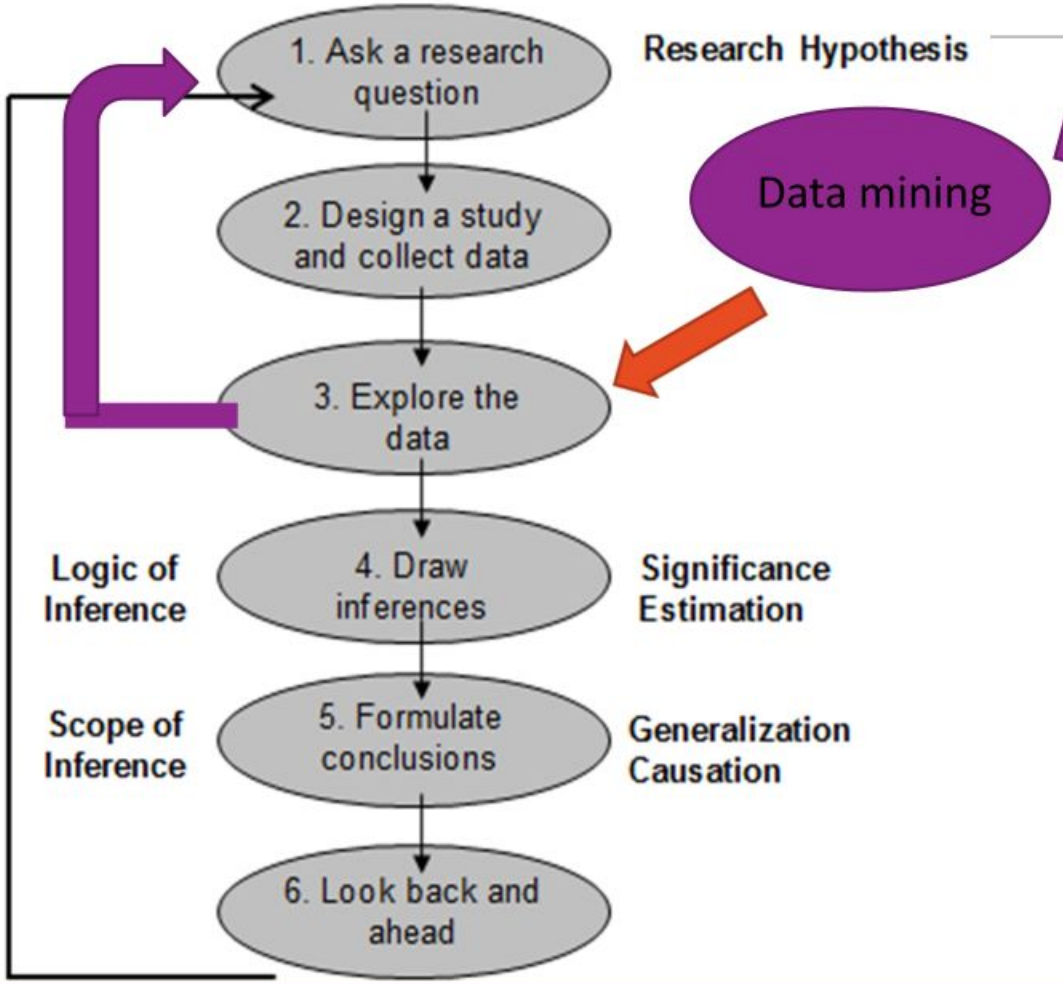
Week 5 Overview

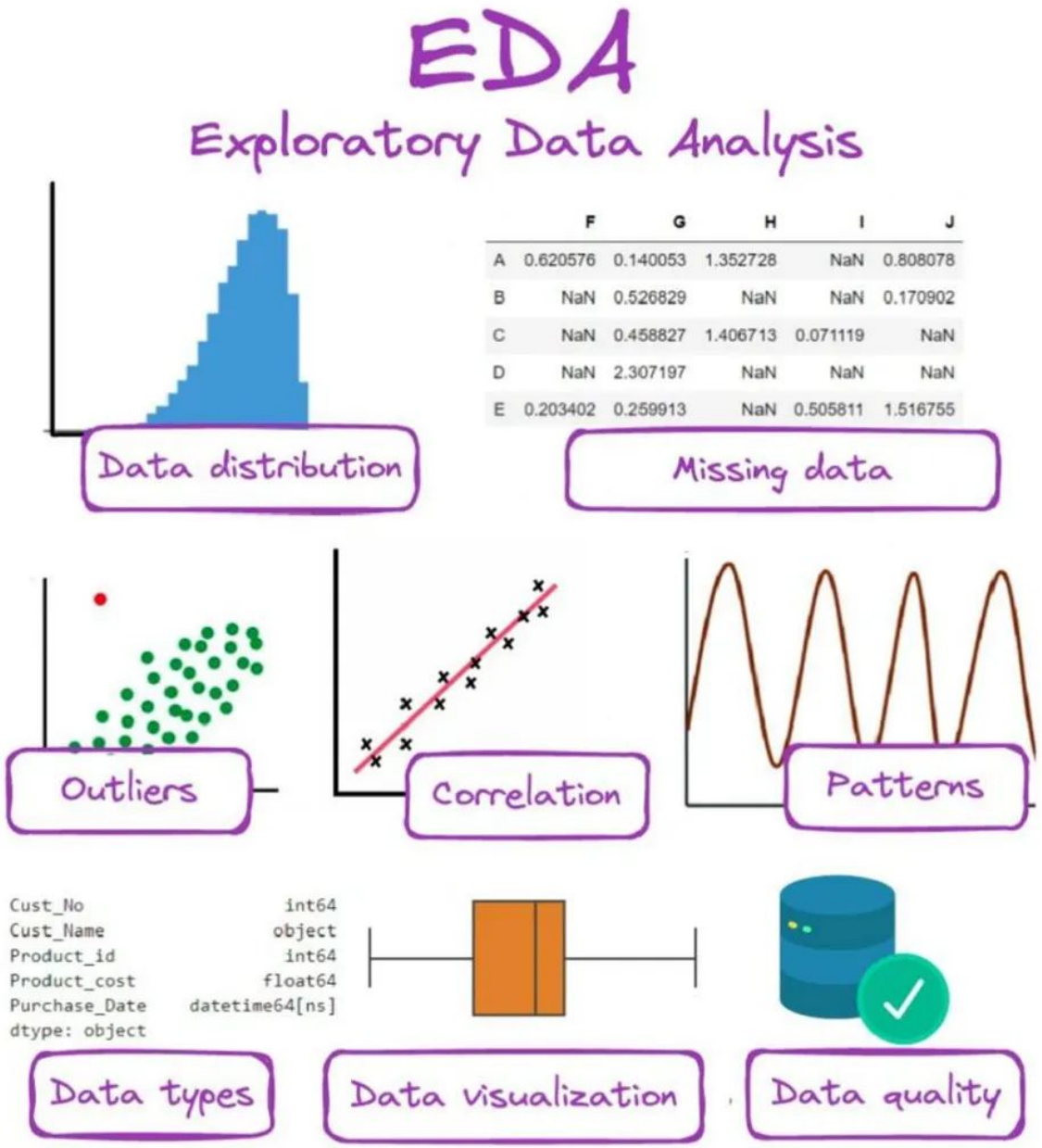
Class A Topic: Resampling methods for non-parametric estimation

Class B Topic: Resampling methods for non-parametric testing

We will introduce a third class of resampling methods, cross-validation, along with methods for model validation.

Eight-Step Process with Big Data





	F	G	H	I	J
A	0.620576	0.140053	1.352728	NaN	0.808078
B	NaN	0.526829	NaN	NaN	0.170902
C	NaN	0.458827	1.406713	0.071119	NaN
D	NaN	2.307197	NaN	NaN	NaN
E	0.203402	0.259913	NaN	0.505811	1.516755

Descriptive and exploratory questions through EDA

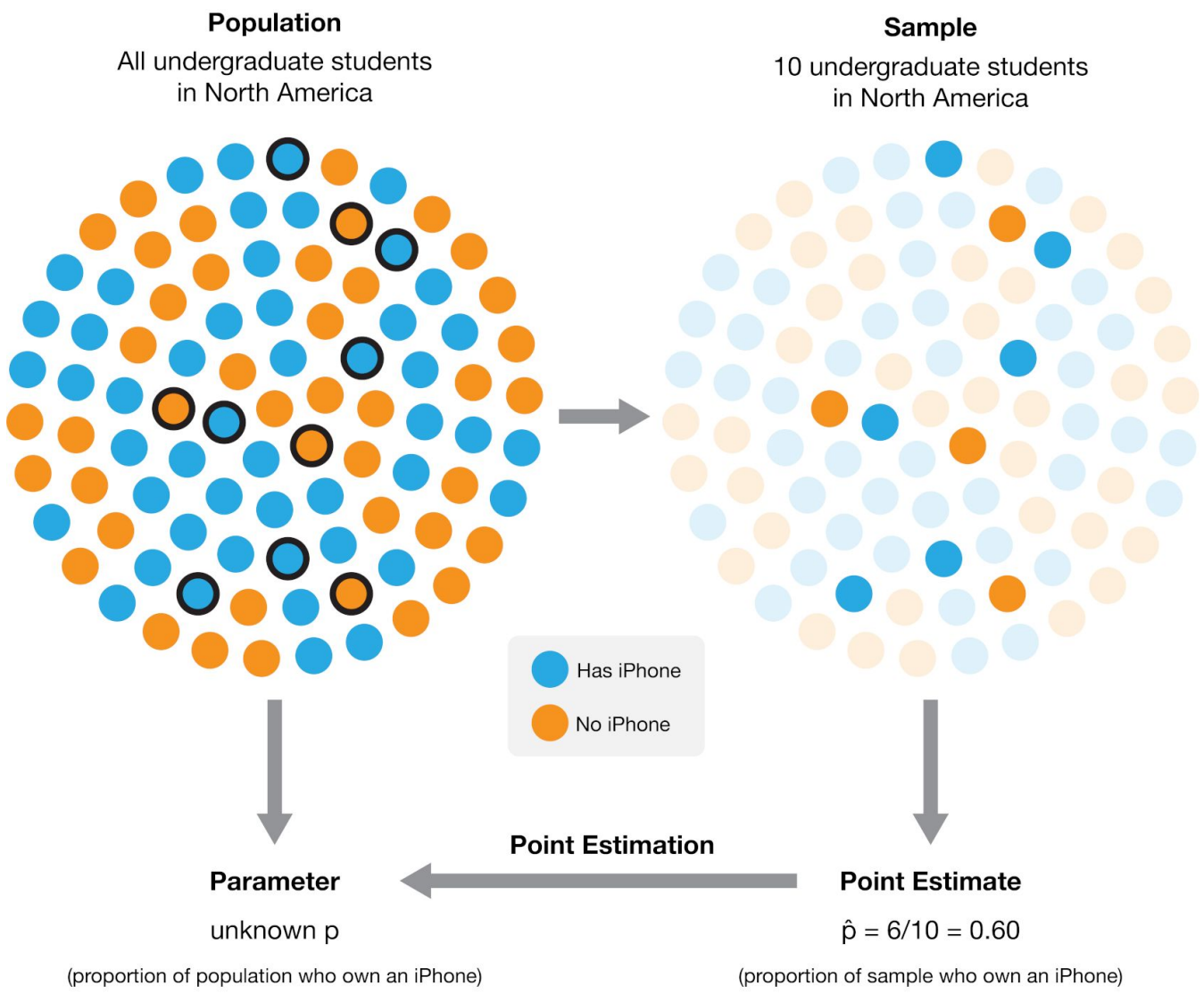
Sets the stage for subsequent statistical analysis or modeling to address:

- Inferential questions
- Predictive questions
- Causal questions
- Mechanistic questions

By using these EDA tools, you can form a strong hypothesis about the true distribution of your data, which then guides your preparation and modeling choices

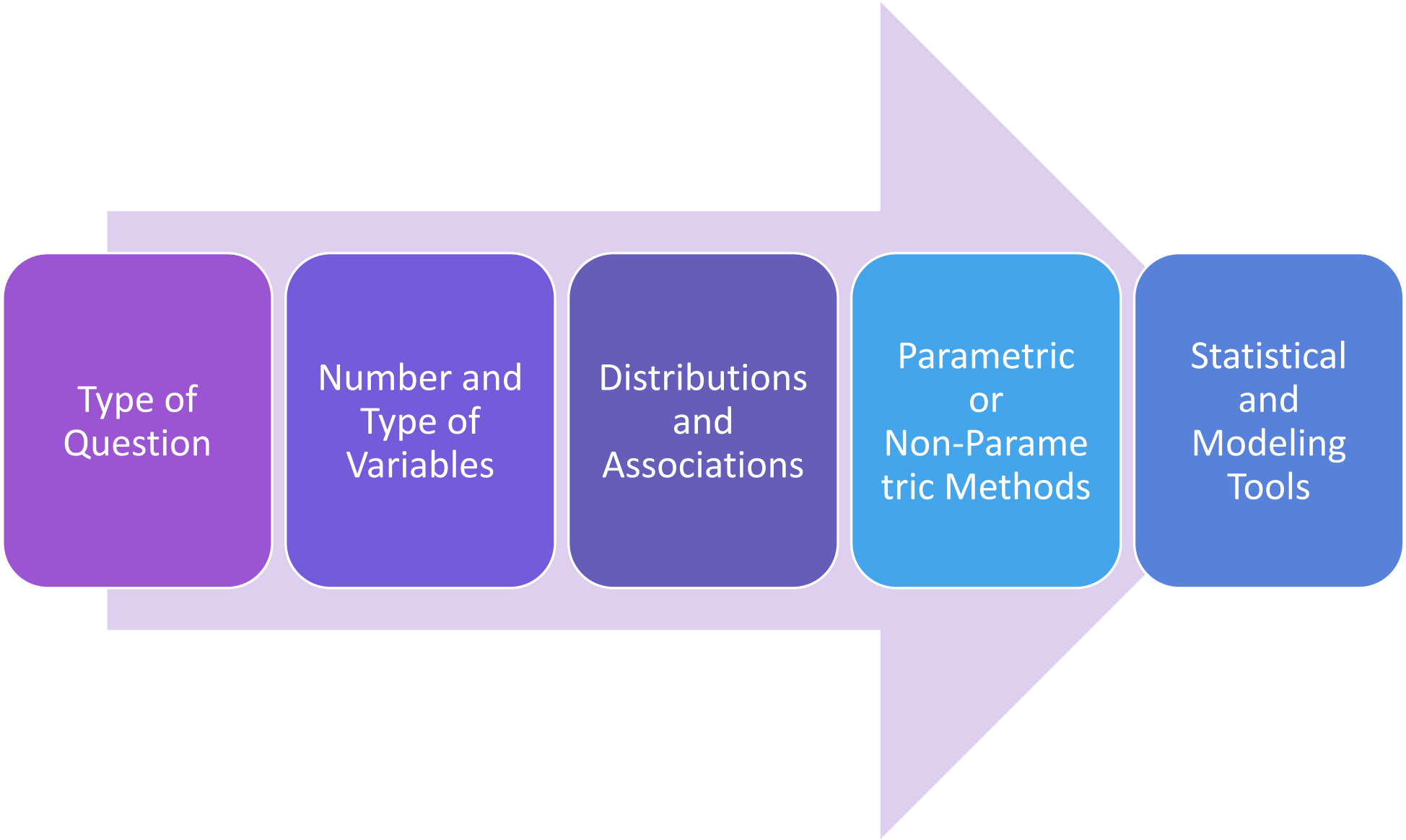
Key Questions for Inference

- **Estimation**: What is the most plausible **value** for a phenomenon ?
- **Significance**: How **strong** is the evidence that this is not just random chance?
- **Generalization**: How **broadly** do the conclusions apply?
- **Causation**: Can we say what **caused** the observed phenomenon?

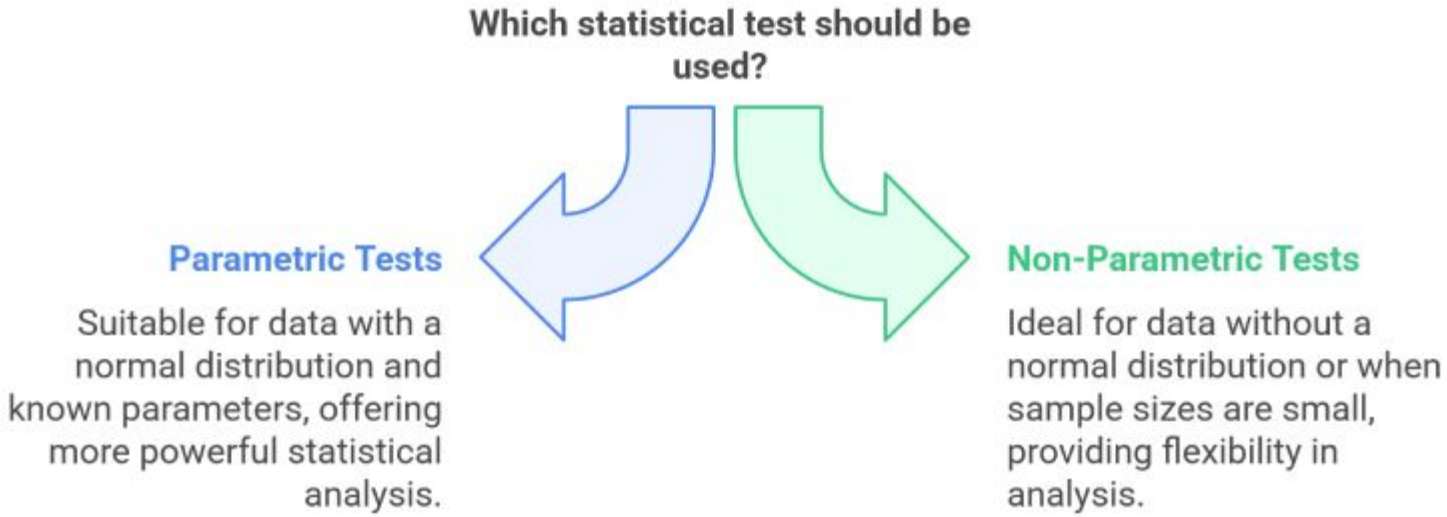

Population
All undergraduate students in North America
Sample
10 undergraduate students in North America
Has iPhone
No iPhone
Point Estimation
Parameter
unknown p
(proportion of population who own an iPhone)
Point Estimate
p̂ = 6/10 = 0.60
(proportion of sample who own an iPhone)

Reminder: Probability Distributions Are Essential for Parametric Inference

- Provide mathematical framework for generalizing from the observed data to the underlying population.
- Most commonly used to find a Maximum Likelihood Estimation (frequentist approach)
  - "Given the data we have observed, which probability distribution parameters make this specific data most likely to have occurred."
  - Assume a distribution and its unknown parameter
  - The probability density (or mass) function becomes the "Likelihood Function"
  - Maximize the likelihood: use calculus or optimization algorithms to find the values of the parameters that maximize the value of the Likelihood function.
- In summary: The distribution provides the blueprint (the likelihood function) and MLE finds the specific dimensions (the parameters) that best fit your data into that blueprint.
- We will try this and Bayesian approaches in a few week.


Type of Question
Number and Type of Variables
Distributions and Associations
Parametric or Non-Parametric Methods
Statistical and Modeling Tools

Parametric Vs. Non-Parametric Tests



https://www.6sigma.us

Parametric or Nonparametric Statistical Tests: Choosing the Most Appropriate Option for your Data

Does the data follow a *normal distribution*?

Are the sample *variances equal*?

Is the data measured on an *interval* or *ratio* scale (as opposed to a *nominal* or *ordinal* scale)?

If all 3 are true, ***Parametric Tests*** are appropriate

If 1 or more is *not* true, ***Nonparametric Tests*** may be more appropriate

Overview of Resampling Techniques for Inference

Resampling Techniques for

Estimation: What is the most plausible **value** for a phenomenon ?

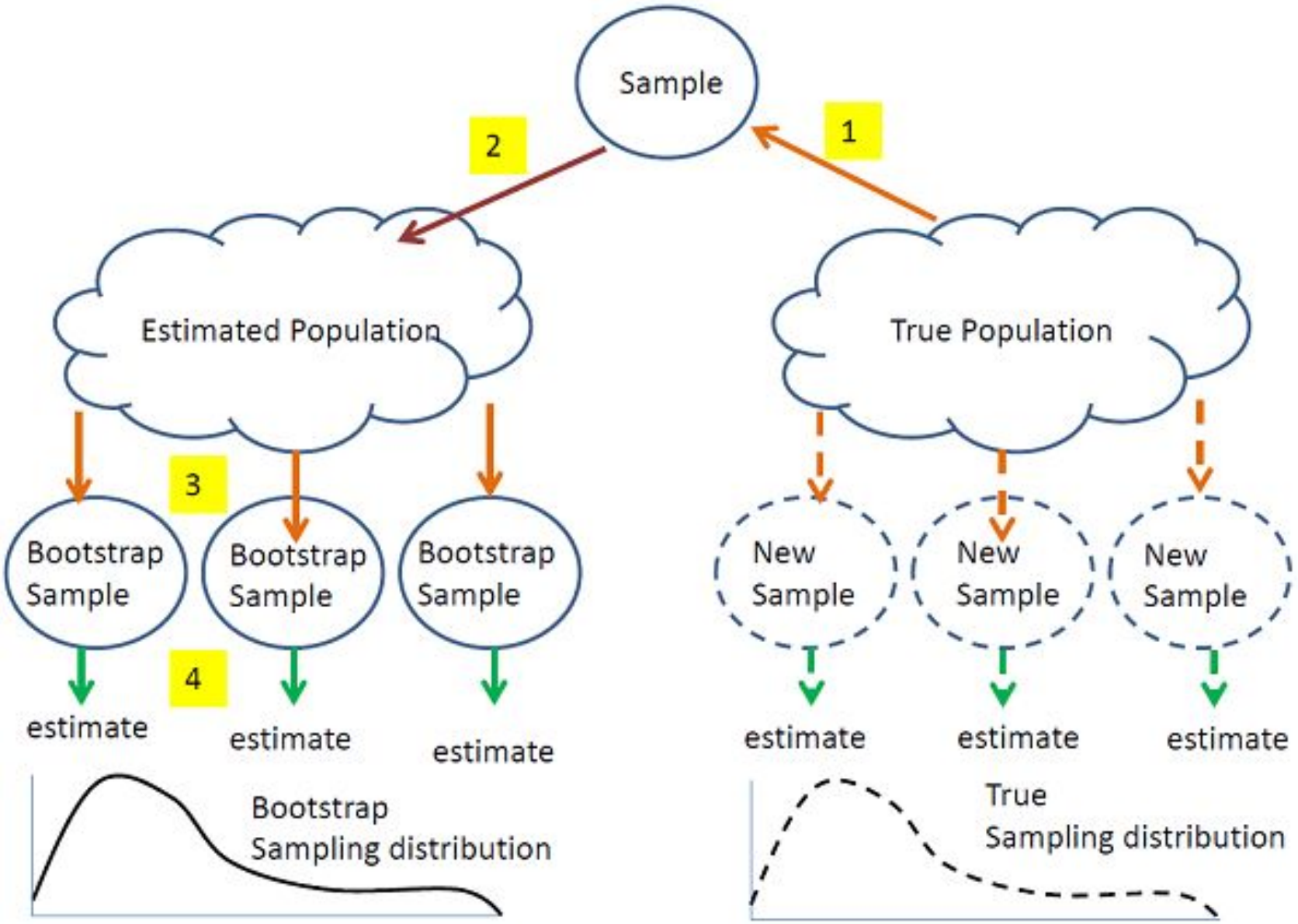
Significance: How **strong** is the evidence that this is not just random chance?

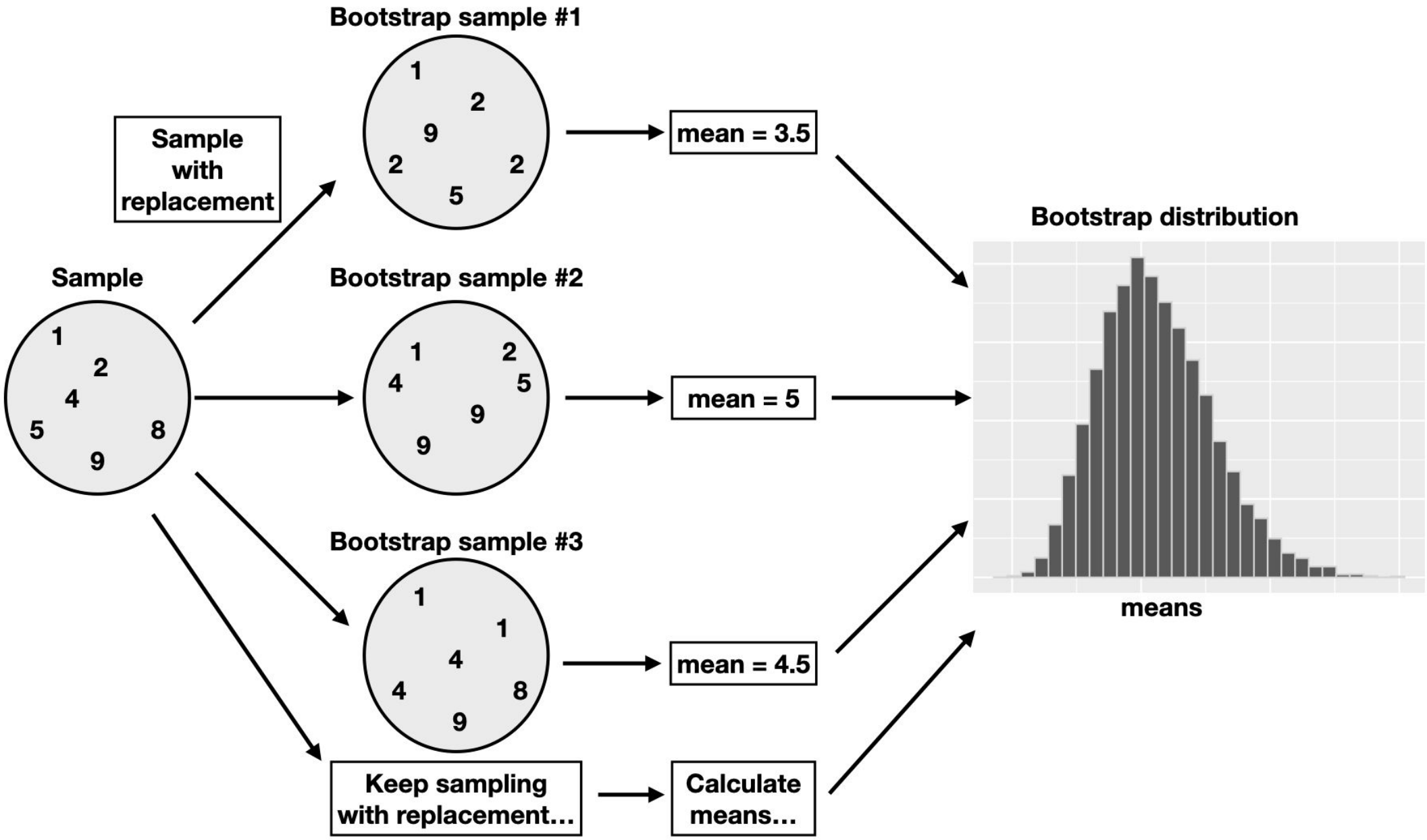
Generalization: How **broadly** do the conclusions apply?

Causation: Can we say what **caused** the observed phenomenon?

Bootstrapping: Alternate Non-Parametric Method

- Data distribution is unknown or non-normal
- Statistic is complex or non-standard
  - Median
  - Ratios, Correlation coefficients, or model coefficients
- When the sample size is small
- When variance and standard error are hard to calculate
- **Summary**: If you're in doubt about assumptions of your favorite parametric test, or if your statistic is unusual, bootstrapping is your nonparametric fallback.


Sample
1
2
Estimated Population
True Population
3
Bootstrap Sample
Bootstrap Sample
Bootstrap Sample
New Sample
New Sample
New Sample
4
estimate
estimate
estimate
estimate
estimate
estimate
Bootstrap Sampling distribution
True Sampling distribution


Bootstrap sample #1
1
2
9
2
2
5
Sample
with
replacement
mean = 3.5
Bootstrap distribution
Sample
1
2
4
5
8
9
Bootstrap sample #2
1
2
4
5
9
9
mean = 5
Bootstrap sample #3
1
1
4
4
8
9
mean = 4.5
means
Keep sampling
with replacement…
Calculate
means…

Why Use Bootstrapping?

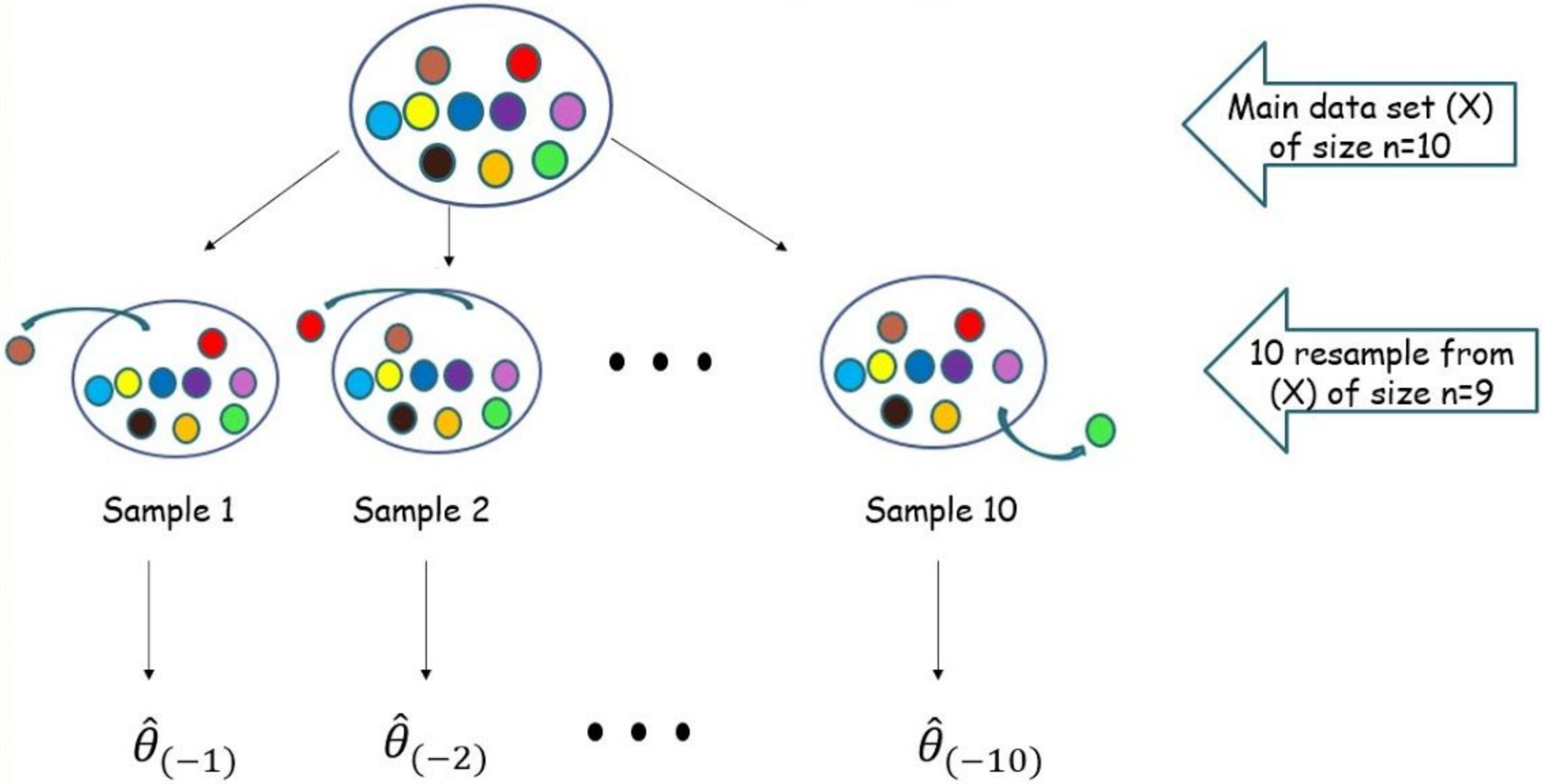
- Better, consistent estimation of variability of sample statistic
- No assumptions about the distribution of your data
  - Use whatever sampling distribution emerges, even if CLT doesn't apply
  - (Same is true for simulating the null distribution)
- Small samples can be usable
- No need for equal variances when comparing groups
- Can be applied to MANY sample statistics, even those without mathematical approaches
  - Mean, median, mode, standard deviation, analysis of variance, correlations, regression coefficients, proportions, odds ratios, variance in binary data, and multivariate statistics

Week 4 Demo Notebook

- Part 1: Bootstrap: Randomly sampling with replacement to create many different datasets
- Active Learning Exercise at end of Part 1:
  - Repeat the analysis with:
    - 10% of the population used as the sample
    - Estimating the medina rather than the mean.

    - https://colab.research.google.com/drive/19tbIDNNJ-gKwOlkikfuFSZBGq1MpFbMI?usp=sharing

Jackknife resampling



Jackknife Demo – Part 2 of WEEK 4A Demo Notebook

Week 5B: Resampling for Hypothesis Testing

Resampling Techniques for

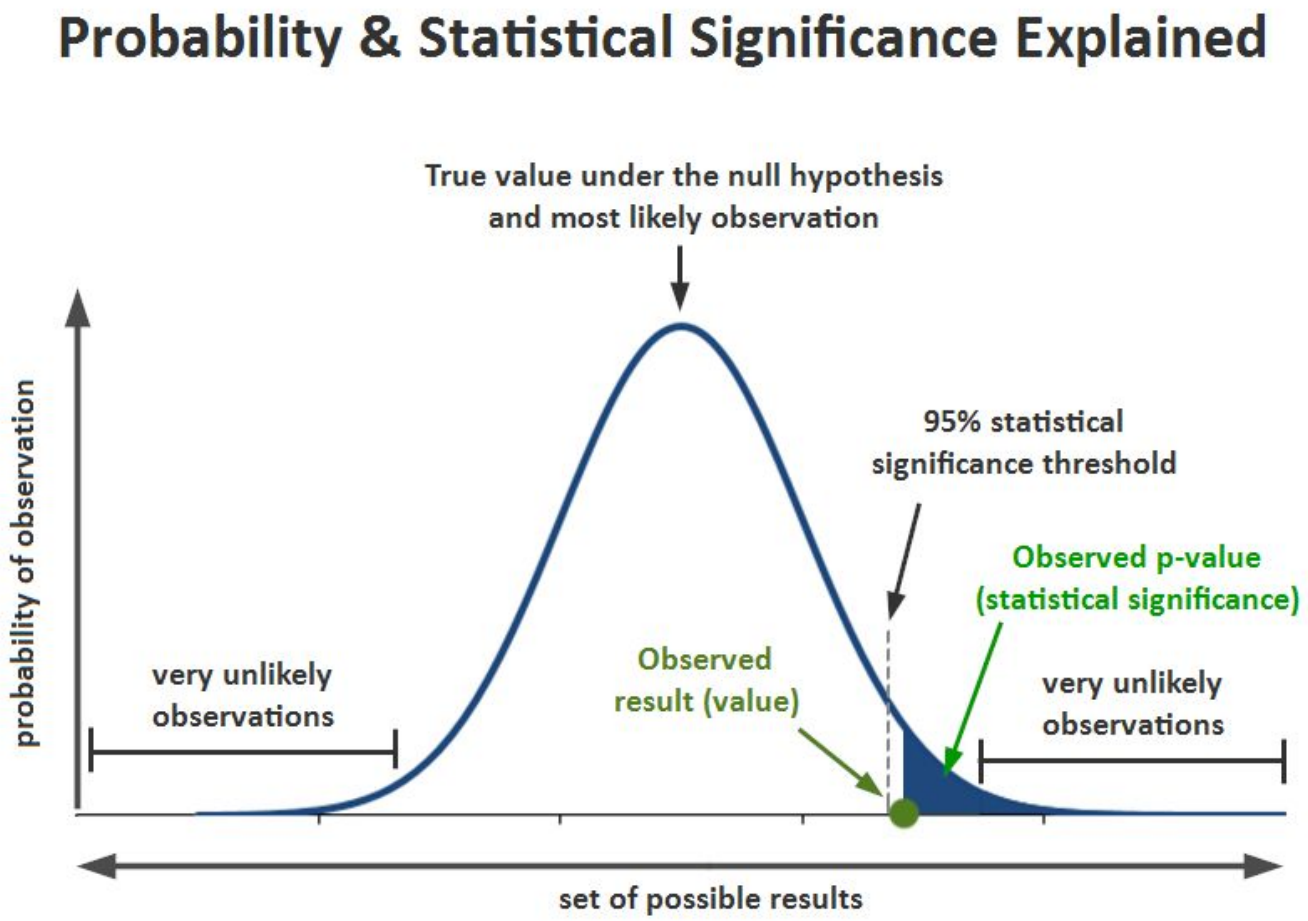
- **Estimation**: What is the most plausible **value** for a phenomenon ?
- **Significance**: How **strong** is the evidence that this is not just random chance?
- **Generalization**: How **broadly** do the conclusions apply?
- **Causation**: Can we say what **caused** the observed phenomenon?

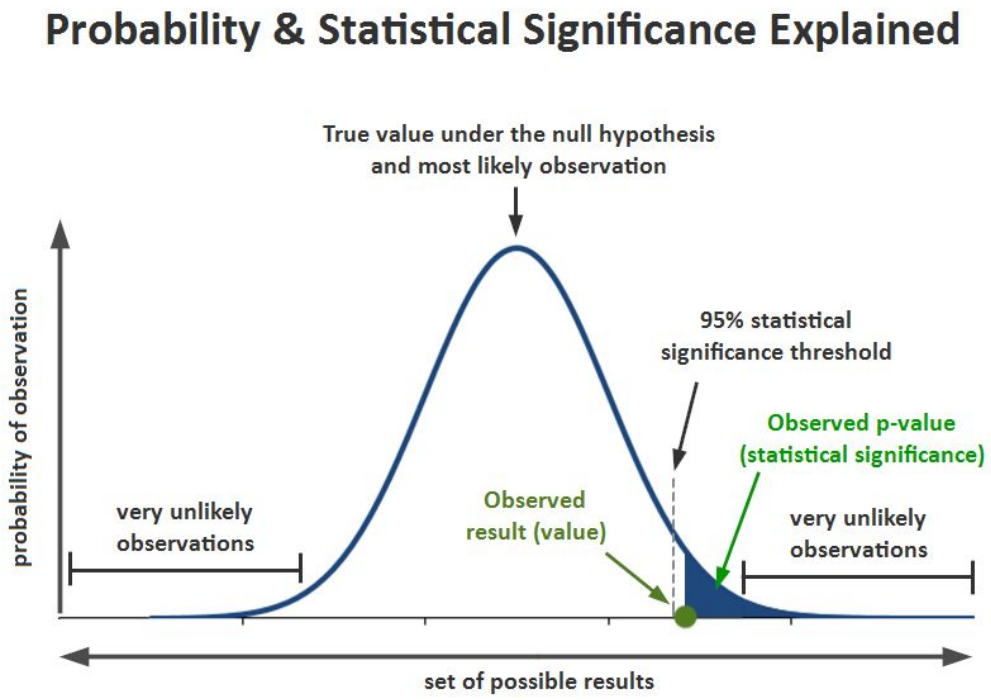
Basic Logic of Inference

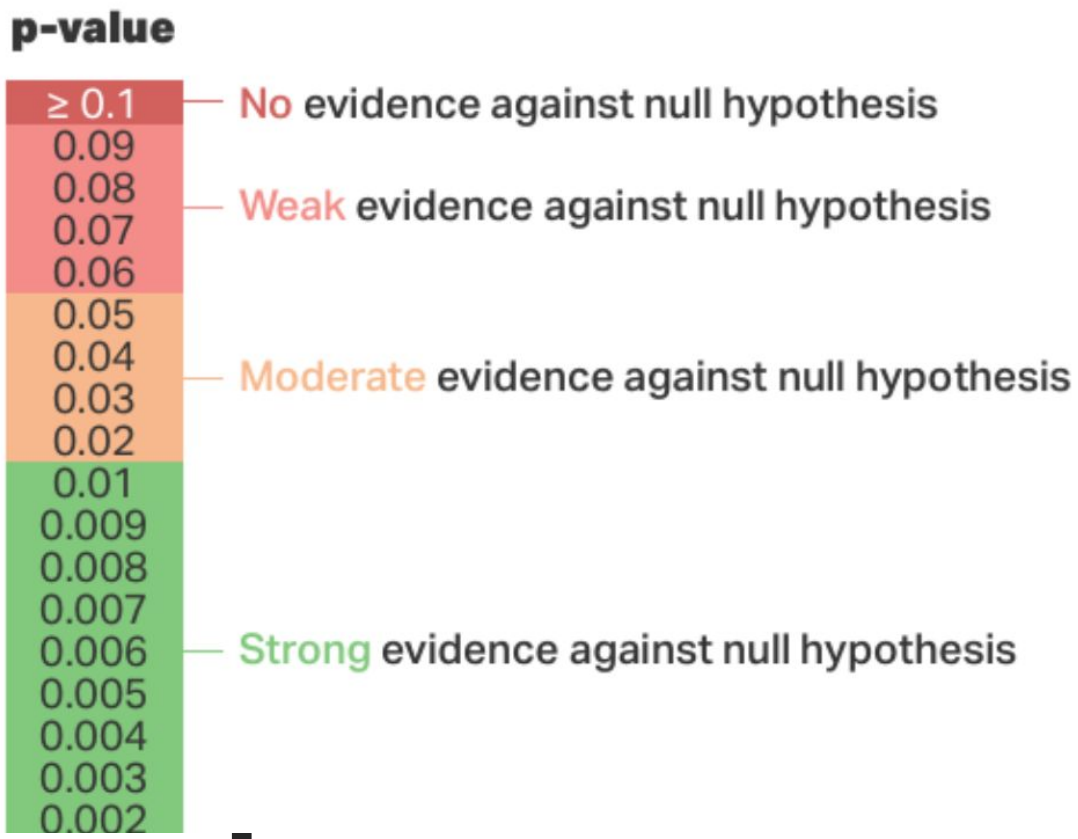
- We have claim, question, or effect that we want to research. [hypothesis]
- We conduct a study or experiment and get some sample results. [statistic]
- We test to see if the sample result is consistent with the random variation we see in the status quo, or if it is more unusual. [simulate]
- We quantify the probability of our sample result, assuming the status quo . [strength]
- We estimate the range of most likely values [*values*]

Null Distribution and Observed Value

- Centered on the null hypothesis (no difference or no effect)
- Distribution of all sample differences, IF the null hypothesis is true.
- If the observed value falls outside the central 95%, we have an unlikely observation
- Reject the null hypothesis if outside the central 95%



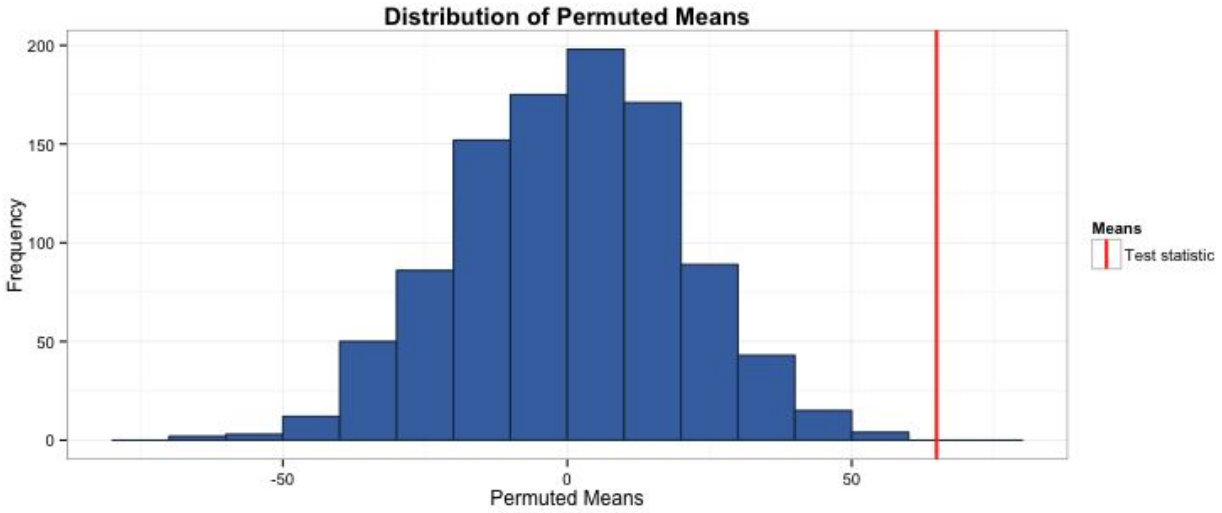


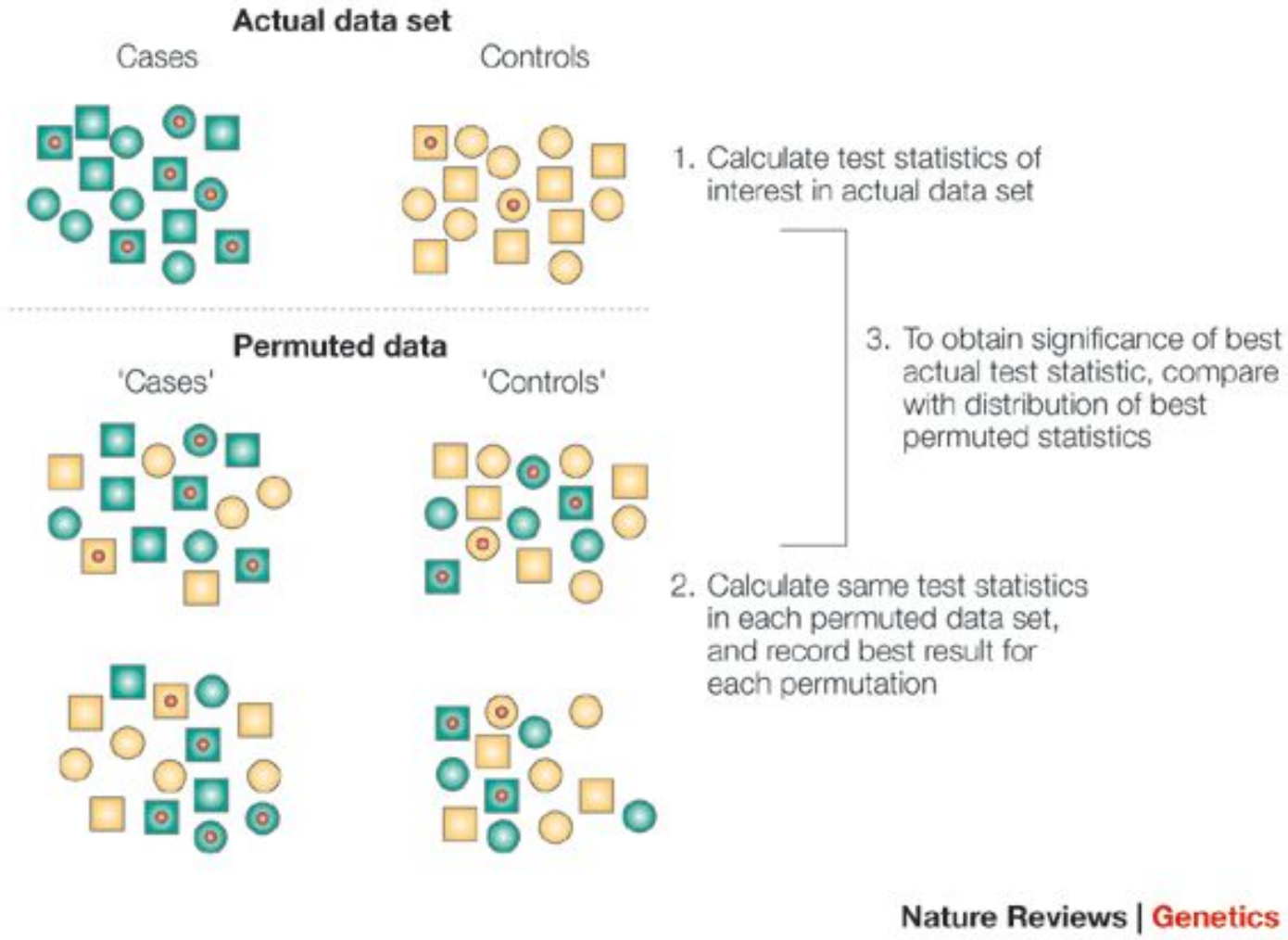


Interpreting the p-value

Permutation tests simulate the null distribution

- Determine whether there is a difference between two groups from the same population (2 samples)
- Simulates the null distribution by randomly reassigning data labels, under the assumption that there is no difference between the groups
- Provides means to determine the probability of obtaining the observed result, if the null hypothesis is true




Actual data set
Cases
Controls
1. Calculate test statistics of interest in actual data set
Permuted data
'Cases'
'Controls'
3. To obtain significance of best actual test statistic, compare with distribution of best permuted statistics
2. Calculate same test statistics in each permuted data set, and record best result for each permutation
Nature Reviews | Genetics

Logic of Permutation Test

- Choose the test statistic (difference of means, difference of medians, etc.)
- State the null hypothesis
- Calculate the observed statistic
- Simulate the null distribution
  - Combine both groups into one data set
  - Shuffle and re-assign labels, keep the same number of each label in each permutation
  - Calculate the test statistic for the permutation
  - Repeat to generate a distribution of test statistics that would occur, if the null hypothesis is true.
- Find the 95% confidence interval of the simulated distribution
- Find the p-value of the observed statistic
- Make a conclusion

Hands- On

DEMO

GROUP ACTIVITY